

PhD Opportunity in the UCL Atmospheric Composition and Air Quality group:

Modelling and measuring pollution plumes from rocket launches

We are pleased to advertise an exciting PhD studentship commencing February 2027 or soon thereafter in the UCL Air Quality and Atmospheric Composition research group supervised by Prof Eloise Marais and funded by a Leverhulme Research Project Grant awarded to Prof Marais.

Details of the opportunity, eligibility, and how to apply are included below.

The research topic:

Join a team of researchers based at UCL, with collaborators at MIT, investigating the ability to use instruments in space to monitor and quantify pollution from rocket launches. Rocket launches are a fast-emerging threat to the atmosphere, releasing harmful pollution into every layer of the atmosphere when these vehicles deliver satellites to orbit. The successful candidate will write computer code to model pollution plumes from rocket launches to investigate the skill of remote sensing and emissions inversion methods to infer how much pollution is produced by rockets.

Eligibility:

Applicants will be considered that hold a First or Upper Second Class honours degree (or equivalent Master's qualification) in a relevant research field, such as Atmospheric Science, Environmental Science, Earth Science, Computer Science, Physical Geography, Chemistry, Mathematics, Physics, Astronomy, Aerospace, Astrophysics. This list is not exclusive. Other relevant disciplines may be suitable. If unsure, please email Prof Marais (e.marais@ucl.ac.uk). No prior experience in remote sensing, emissions inversions or plume modelling is required.

What the studentship covers:

The studentship includes a monthly stipend for 3.5 years and covers tuition fees at the Home (UK) rate. The initial annual stipend of £23,805 for 26/27 academic year is awarded pro rata. An indication of the stipend for subsequent years is provided on the UCL studentships budget page ([link to webpage](#)).

The studentship covers tuition fees at the UK (Home) rate. Eligibility for UK rates is explained here ([link to website](#)). International students wishing to apply for the studentship would need to source other funding to cover the difference between the overseas and home tuition fee.

How to apply:

To apply, follow the submission instructions on this page: <https://www.ucl.ac.uk/prospective-students/graduate/apply>. When prompted for a PhD programme, select the **RRDGEOSING01** "full time" option. One of the requirements is to submit a brief "Project proposal". As the PhD project is pre-determined, instead detail in 400-500 words your interest and suitability in pursuing a PhD and in working on the stated project.

Notify Prof Marais via email (e.marais@ucl.ac.uk) after you have submitted your application.

Closing date:

The closing date for applications is **Thursday 19 November 2026**. Shortlisted candidates will be selected to proceed to a brief interview. Details and the timing of the interview will be sent to shortlisted candidates soon after the application deadline.

Additional Information:

- Links to relevant websites:
Prof Marais institution profile: <https://profiles.ucl.ac.uk/77783-elioise-marais>
Marais research group website: <https://maraisresearchgroup.co.uk/>
Orbital emissions tracker: <https://orbitalemissions.org/>
- For questions about the research project, your suitability, or the academic aspects of the studentship, please email Prof Marais (e.marais@ucl.ac.uk).
- For questions about the application process or if you encounter difficulties with your application, please contact the SHS Postgraduate Team at shs.pgr@ucl.ac.uk, using the subject "Query about Prof Marais' PhD studentship".